



Financial Performance Analysis



Capstone Project
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Introduction

In the modern business landscape, organizations generate and manage vast volumes of financial data from multiple sources such as transactions, customers, vendors, and employees. Analyzing this data effectively is essential for understanding business performance and supporting strategic growth. By combining these datasets, the project aims to develop a comprehensive view of business operations and financial performance. The analysis helps uncover meaningful patterns, improve operational efficiency, and support data-driven decision-making processes.

Project Objectives

- To analyze financial and operational data from multiple business sources.
- To gain insights into customer behavior, vendor performance, and employee-related expenses.
- To evaluate revenue, expenditure, and overall profitability trends.
- To identify patterns and opportunities that can improve business efficiency.
- To support better strategic and financial decision-making through meaningful data insights.
- To create a structured analytical approach for monitoring organizational performance.



Data Cleaning & Validation Process

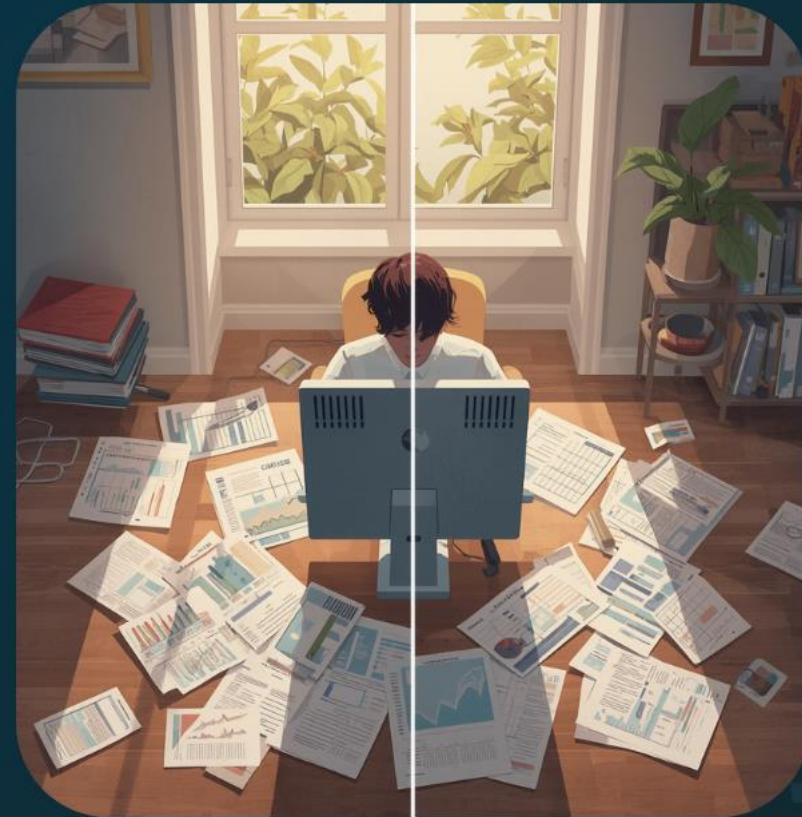
- To ensure reliable and high-quality analysis, a comprehensive data cleaning and validation process was carried out on a database containing more than 10,400 raw financial transactions. The primary objective of this phase was to improve data accuracy, consistency, and integrity before performing analytical operations.
- Several preprocessing techniques were applied to eliminate inconsistencies and prepare the dataset for meaningful insights and visualization.

Essential Cleanup Steps

- Fixing Broken Date Formats
- Removing Errors & Inconsistencies
- Aligning Financial Data Formats
- Final Data Verification

Handling Nullable Foreign Keys

- The Financial_Transactions dataset included nullable foreign key fields such as customer_id and vendor_id.
- customer_id remains nullable for non-revenue transactions
- vendor_id remains nullable for non-expense transactions





Technical Implementation (The Python Pipeline)

Tools & Technologies Used

- Python was used as the primary programming language for data processing and automation.
- pandas was utilized for data cleaning, transformation, filtering, and dataset merging operations.
- numpy was used for handling numerical computations and optimizing data-processing tasks.

Data Processing & Automation Workflow

- **Standardizing Date Formats**
The raw datasets contained inconsistent date formats, particularly inverted day/month values. Python functions such as `to_datetime()` with `dayfirst=True` were implemented to automatically standardize date structures, ensuring accurate time-based analysis and reporting.
- **Integrating Multiple Datasets**
Automated scripts were written to establish relationships between customer records, vendor datasets, and financial transaction tables. Matching ID columns were used to merge datasets accurately and maintain relational consistency across the database.
- **Data Validation & Transformation**
Additional transformation and validation logic was applied to handle missing values, ensure consistent formatting, and optimize the datasets for analytical processing. This helped improve data reliability and reporting accuracy.





Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to study financial trends, revenue growth, expenses, and overall business performance across a 24-month period. Using Python-based analysis and visualizations, we examined relationships between sales, operational costs, customer transactions, vendor spending, and budget utilization to identify important financial patterns and business insights.

Monthly expense data was aggregated and matched with budget datasets to calculate:

- Expense Variance
- Budget Utilization Percentage
- Category-wise Cost Distribution

This analysis highlighted areas where operational spending exceeded planned budgets and helped detect cost spikes across specific categories and departments.



Statistical Analysis & Hypothesis Testing

To understand whether the company's increasing expenses were temporary fluctuations or part of a recurring financial issue, detailed statistical analysis and hypothesis testing techniques were applied to the financial datasets. Methods such as ANOVA (Analysis of Variance) and Independent T-Tests were performed across multiple expense categories, business units, and vendor transactions to identify significant variations in spending behavior and budget performance.

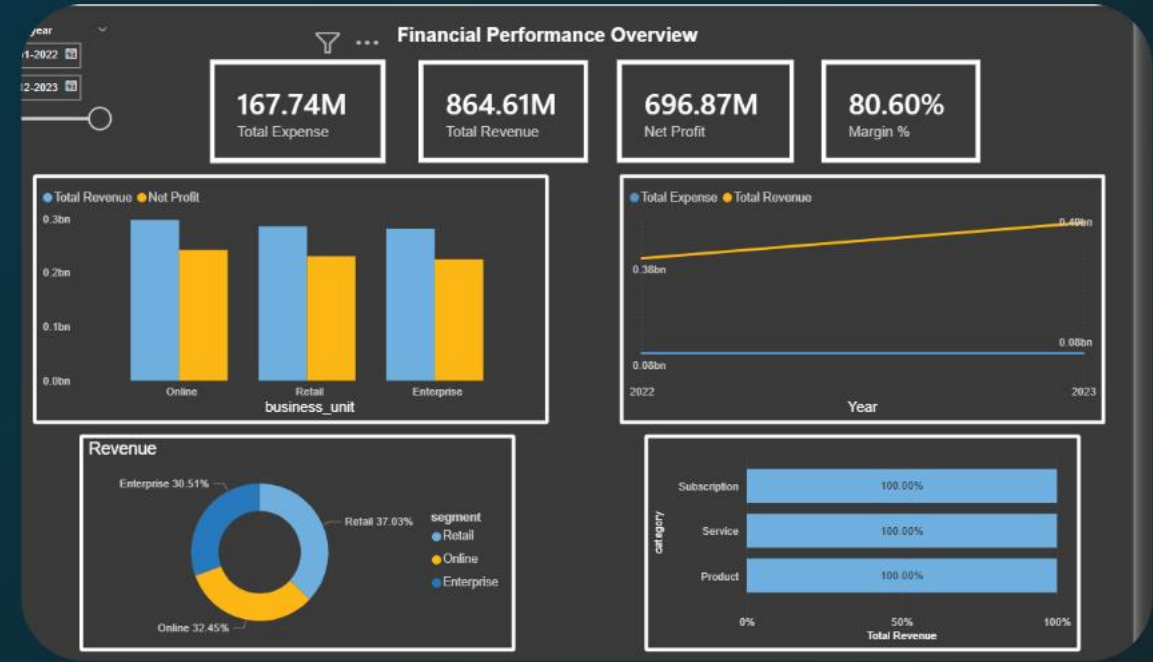
The statistical results showed p-values significantly below the threshold value of 0.05, mathematically confirming that the overspending trends were statistically significant and not caused by random chance. The analysis further identified April 2023 as a major outlier month with unusually high operational expenses that heavily impacted annual financial performance.



Dashboard Development (Power BI)

The final stage of the project involved transforming the cleaned datasets, statistical findings, and business insights into a fully interactive Power BI dashboard designed for efficient financial monitoring and executive-level decision-making.

The dashboard was built using an optimized relational data model, where multiple tables were connected through automated relationships and custom DAX (Data Analysis Expressions) measures. This enabled fast calculations, dynamic filtering, drill-down analysis, and real-time interaction between different business metrics and visual components.



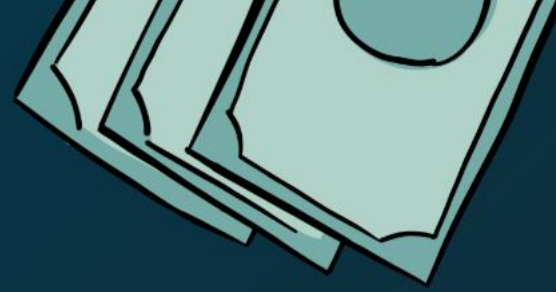
Key Insights

- The company generated a total revenue of \$864.61 Million with an impressive 80.60% Net Profit Margin, resulting in approximately \$696.87 Million in net profit.
- Financial trends showed consistent revenue growth across the 24-month analysis period, indicating strong business performance and profitability.
- Statistical testing confirmed that spending patterns and expense fluctuations were statistically significant rather than random financial variations.

Analytical Findings

- The organization experienced a major 60.88% budget overrun, with almost every business unit exceeding its planned financial limits.
- Expense analysis identified Marketing (17.26%) and Supplies (16.54%) as the highest spending categories contributing to cost escalation.
- Statistical analysis and outlier detection highlighted April 2023 as an abnormal high-spending period that significantly impacted annual financial performance.
- Vendor analysis revealed heavy dependency on multi-purpose vendors such as Boyd, Lozano, and Gomez, who were billing across multiple departments and expense categories, increasing operational and procurement costs.
- The findings emphasized the need for stronger budget control mechanisms, vendor diversification strategies, and improved financial governance across the organization.



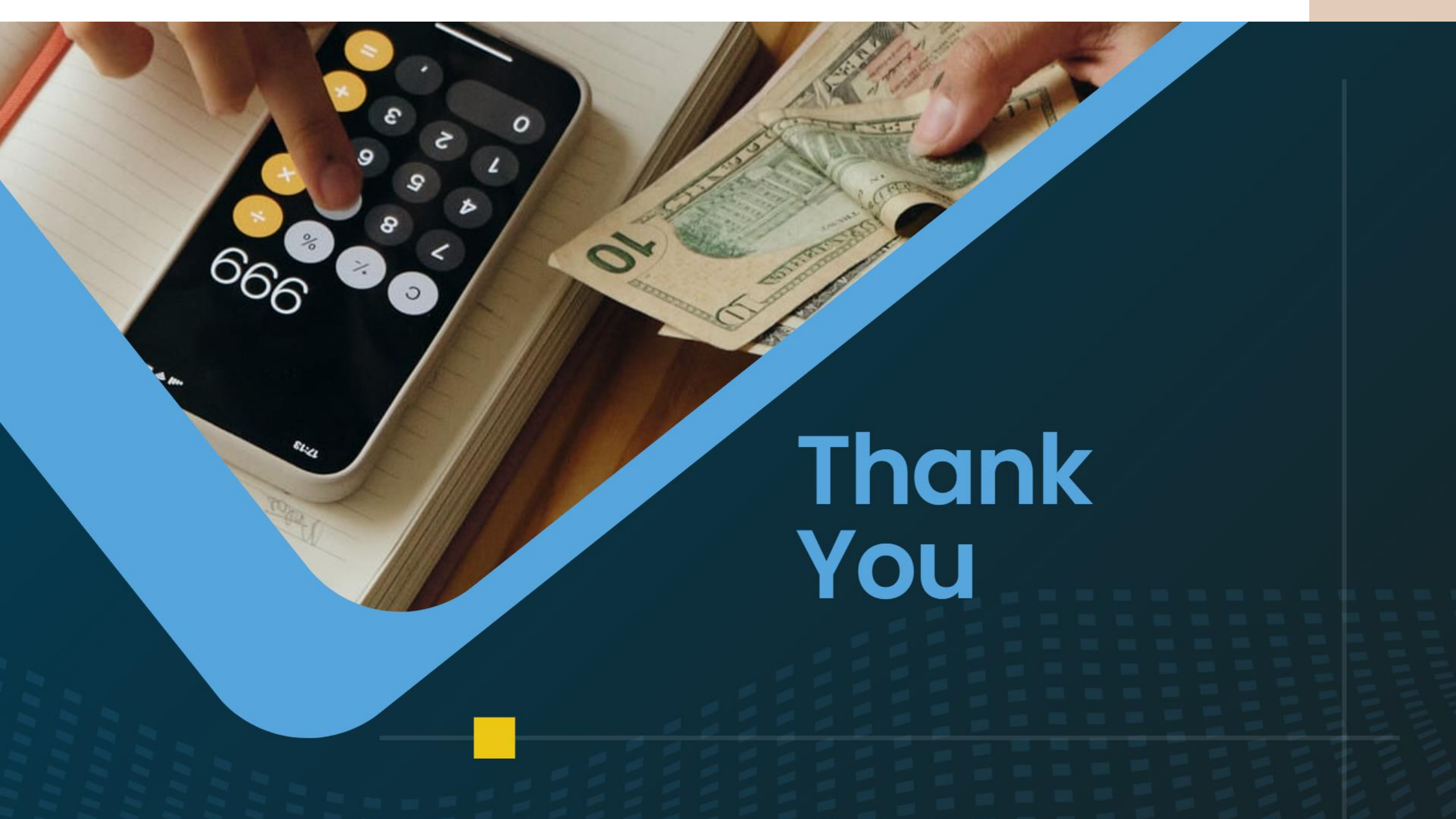


Strategic Summary

The analysis revealed that while the company is achieving exceptionally high profitability and strong business growth, it is simultaneously facing major challenges in budget control, operational spending, and vendor dependency, highlighting the need for improved financial governance, optimized cost management, and more strategic allocation of resources to sustain long-term growth efficiently.

Strategic Recommendations

- Redesign the existing budget structure to better align with current business growth, especially within rapidly expanding online and digital business segments.
- Conduct vendor audits and negotiate long-term bulk discount contracts with high-spend vendors to reduce procurement costs and improve financial efficiency.
- Reallocate investments from underperforming operational areas toward high-performing online and subscription-based channels to maximize profitability and long-term scalability.



**Thank
You**

